

Solution Requirements

Date	1 July 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Functional and Non-Functional Requirements:

Step 1: Functional Requirements (FR)

Requirement ID	Requirement Name	Requirement Description
FR-01	User Registration/Login	The system should allow users to register and securely log in to the application.
FR-02	Applicant Data Input	The system should allow users to enter applicant details such as income, employment status, education, family status, housing type, and credit history.
FR-03	Data Validation	The system should validate all input fields and display appropriate error messages for invalid or missing data.
FR-04	Data Preprocessing	The system should preprocess the input data by encoding categorical values and applying the same transformations used during model training.
FR-05	Model Loading	The system should load the trained machine learning model before making predictions.
FR-06	Credit Card Approval Prediction	The system should predict whether a credit card application is likely to be Approved or Rejected.
FR-07	Risk Assessment	The system should identify high-risk applicants based on the prediction outcome.
FR-08	Prediction Display	The system should display the prediction result along with a confidence score (if available).
FR-09	Applicant History	The system should store applicant details and prediction results for future reference.
FR-10	Report Generation	The system should allow users to download or print the prediction report.
FR-11	Dashboard	The system should display applicant information and prediction history in a dashboard.
FR-12	Error Handling	The system should display meaningful error messages when prediction fails or invalid data is entered.

Step 2: Non-Functional Requirements (NFR)

Requirement ID	Requirement Name	Requirement Description
NFR-01	Performance	The system should generate prediction results within 2–3 seconds.
NFR-02	Accuracy	The machine learning model should achieve high prediction accuracy on unseen data.
NFR-03	Security	User credentials and applicant data should be securely stored and protected.
NFR-04	Reliability	The system should operate consistently without unexpected failures.
NFR-05	Scalability	The application should support multiple users simultaneously.
NFR-06	Usability	The user interface should be simple, intuitive, and easy to navigate.
NFR-07	Availability	The system should be available whenever users need to make predictions.
NFR-08	Maintainability	The system should support future model updates and feature enhancements.